English

Introduction

Get Started

Contribution Guide

API Reference

Using Flowise

Agentflows

Multi-Agents

Sequential Agents

API

Analytic

Document Stores

Embed

Monitoring

Streaming

Powered by GitBook

Multi-Agents

Learn how to use Multi-Agents in Flowise, written by @toi500

This guide intends to provide an introduction of the multi-agent AI system architecture within Flowise, detailing its components, operational constraints, and workflow.

Concept

Analogous to a team of domain experts collaborating on a complex project, a multi-agent system uses the principle of specialization within artificial intelligence.

This multi-agent system utilizes a hierarchical, sequential workflow, maximizing efficiency and specialization.

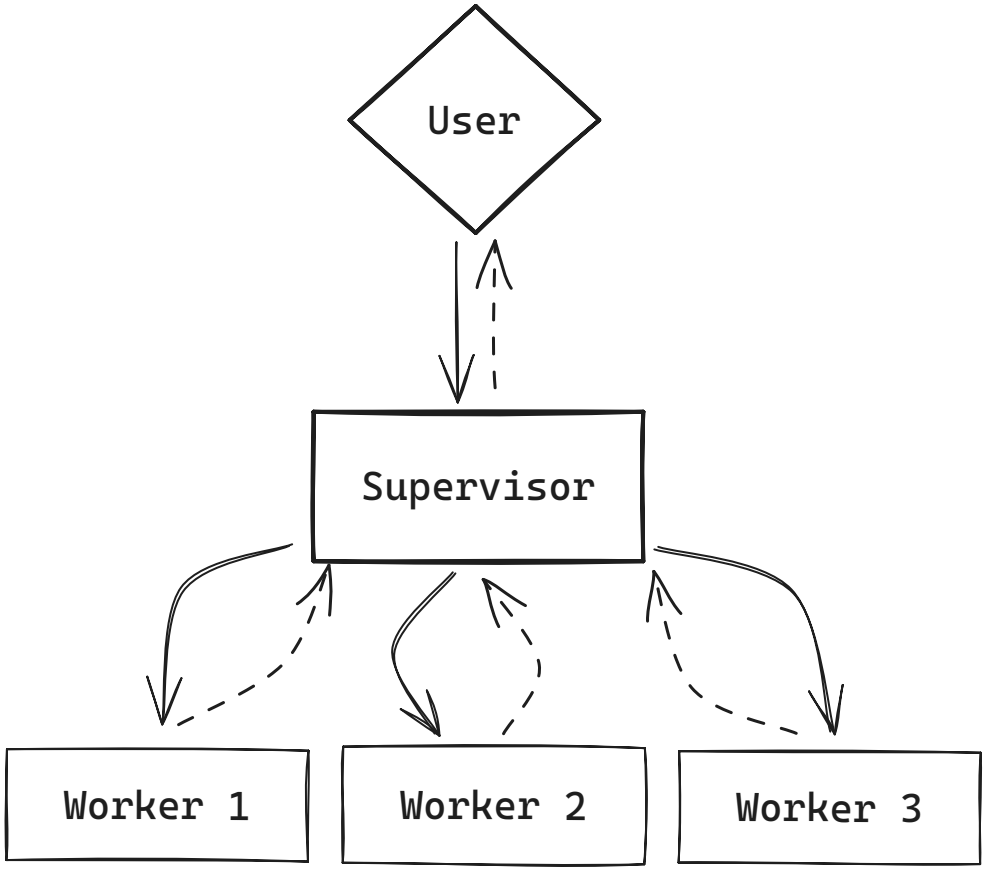
1. System Architecture

We can define the multi-agent AI architecture as a scalable AI system capable of handling complex projects by breaking them down into manageable sub-

tasks.

In Flowise, a multi-agent system comprises two primary nodes or agent types and a user, interacting in a hierarchical graph to process requests and deliver a targeted outcome:

1. **User:** The user acts as the **system's starting point**, providing the initial input or request. While a multi-agent system can be designed to handle a wide range of requests, it's important that these user requests align with the system's intended purpose. Any request falling outside this scope can lead to inaccurate results, unexpected loops, or even system errors. Therefore, user interactions, while flexible, should always align with the system's core functionalities for optimal performance.
2. **Supervisor AI:** The Supervisor acts as the **system's orchestrator**, overseeing the entire workflow. It analyzes user requests, decomposes them into a sequence of sub-tasks, assigns these sub-tasks to the specialized worker agents, aggregates the results, and ultimately presents the processed output back to the user.
3. **Worker AI Team:** This team consists of specialized AI agents, or Workers, each instructed - via prompt messages - to handle a specific task within the workflow. These Workers operate independently, receiving instructions and data from the Supervisor, **executing their specialized functions**, using tools as needed, and returning the results to the Supervisor.



2. Operational Constraints

To maintain order and simplicity, this multi-agent system operates under two important constraints:

- **One task at a time:** The Supervisor is intentionally designed to focus on a single task at a time. It waits for the active Worker to complete its task and return the results before it analyzes the next step and delegates the

subsequent task. This ensures each step is completed successfully before moving on, preventing overcomplexity.

- **One Supervisor per flow:** While it's theoretically possible to implement a set of nested multi-agent systems to form a more sophisticated hierarchical structure for highly complex workflows, what LangChain defines as "Hierarchical Agent Teams", with a top-level supervisor and mid-level supervisors managing teams of workers, Flowise's multi-agent systems currently operate with a single Supervisor.

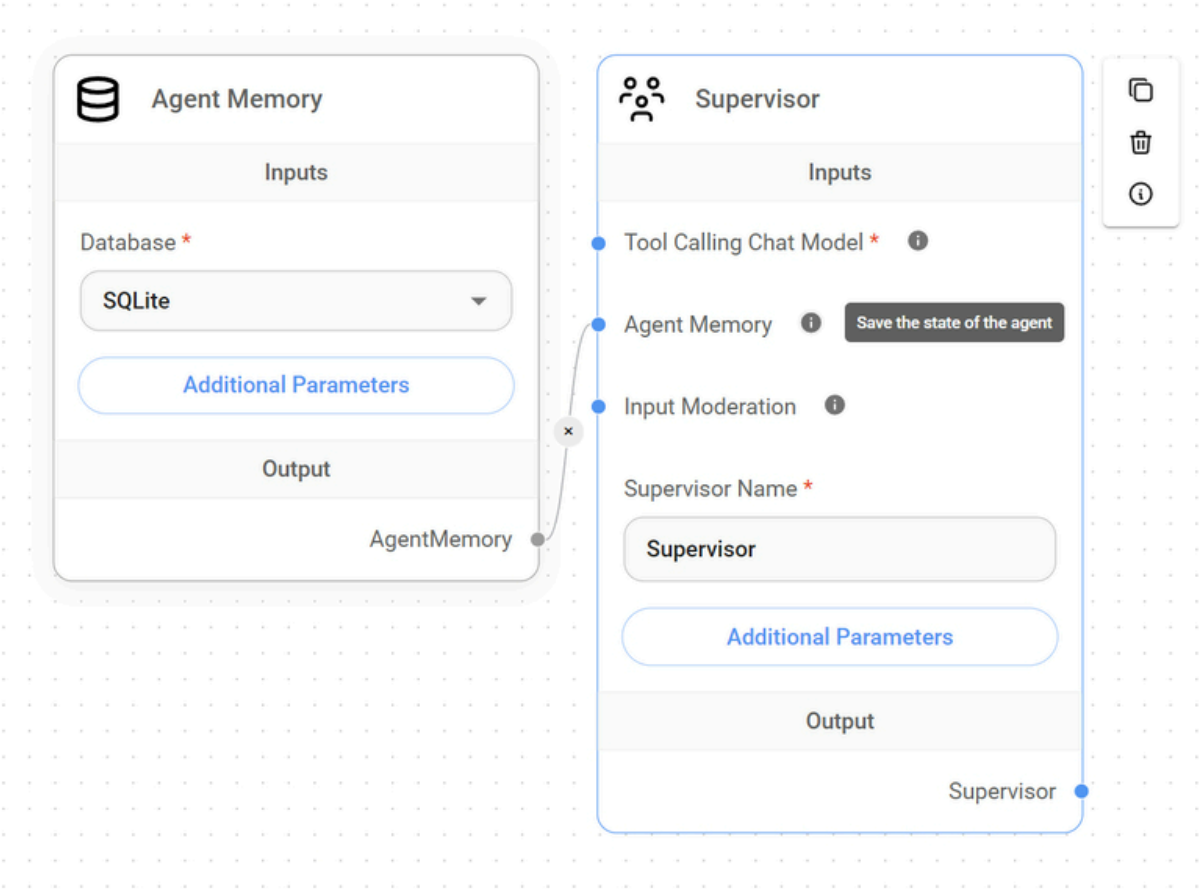
> These two constraints are important when **planning your application's workflow**. If you try to design a workflow where the Supervisor needs to delegate multiple tasks simultaneously, in parallel, the system won't be able to handle it and you'll encounter an error.

The Supervisor

The Supervisor, as the agent governing the overall workflow and responsible for delegating tasks to the appropriate Worker, requires a set of components to function correctly:

- **Chat Model capable of function calling** to manage the complexities of task decomposition, delegation, and result aggregation.
- **Agent Memory (optional)**: While the Supervisor can function without Agent Memory, this node can significantly enhance workflows that require access to past Supervisor states. This **state preservation** could allow the

Supervisor to resume the job from a specific point or leverage past data for improved decision-making.

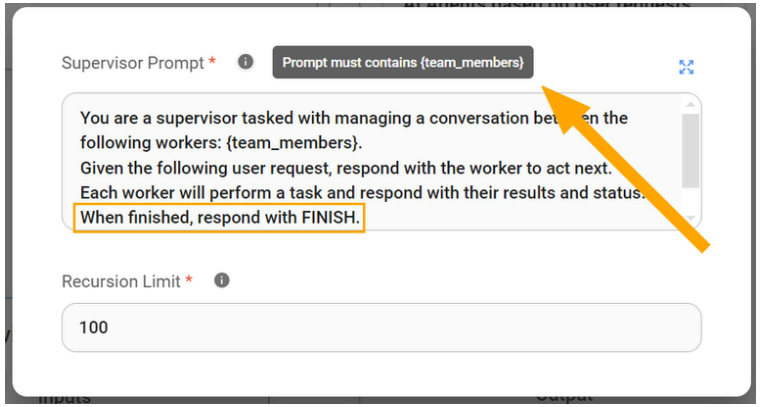


Supervisor Prompt

By default, the Supervisor Prompt is worded in a way that instructs the Supervisor to analyze user requests, decompose them into a sequence of sub-tasks, and assign these sub-tasks to the specialized worker agents.

While the Supervisor Prompt is customizable to fit specific application needs, it always requires the following two key elements:

- **The {team_members} Variable:** This variable is crucial for the Supervisor's understanding of the available workforce since it provides the Supervisor with list of Worker names. This allows the Supervisor to diligently delegate tasks to the most appropriate Worker based on their expertise.
- **The "FINISH" Keyword:** This keyword serves as a signal within the Supervisor Prompt. It indicates when the Supervisor should consider the task complete and present the final output to the user. Without a clear "FINISH" directive, the Supervisor might continue delegating tasks unnecessarily or fail to deliver a coherent and finalized result to the user. It signals that all necessary sub-tasks have been executed and the user's request has been fulfilled.

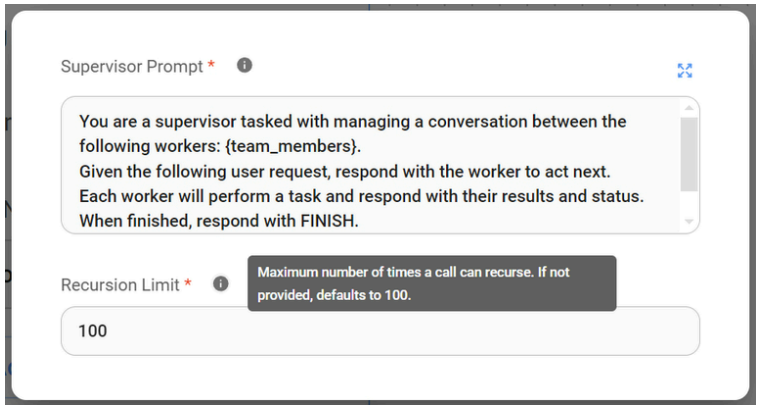


> It's important to understand that the Supervisor plays a very distinct role from Workers. Unlike Workers, which can be tailored with highly specific instructions, the **Supervisor operates most effectively with**

general directives, which allow it to plan and delegate tasks as it deems appropriate. If you're new to multi-agent systems, we recommend sticking with the default Supervisor prompt

Understanding Recursion Limit in Supervisor node:

This parameter restricts the maximum depth of nested function calls within our application. In our current context, **it limits how many times the Supervisor can trigger itself within a single workflow execution**. This is important for preventing unbounded recursion and ensuring resources are used efficiently.



How the Supervisor works

Upon receiving a user query, the Supervisor initiates the workflow by analyzing the request and discerning the user's intended outcome.

Then, leveraging the `{team_members}` variable in the Supervisor Prompt, which only provides a list of available Worker AI names, the Supervisor infers each Worker's specialty and strategically selects the most suitable Worker for each task within the workflow.

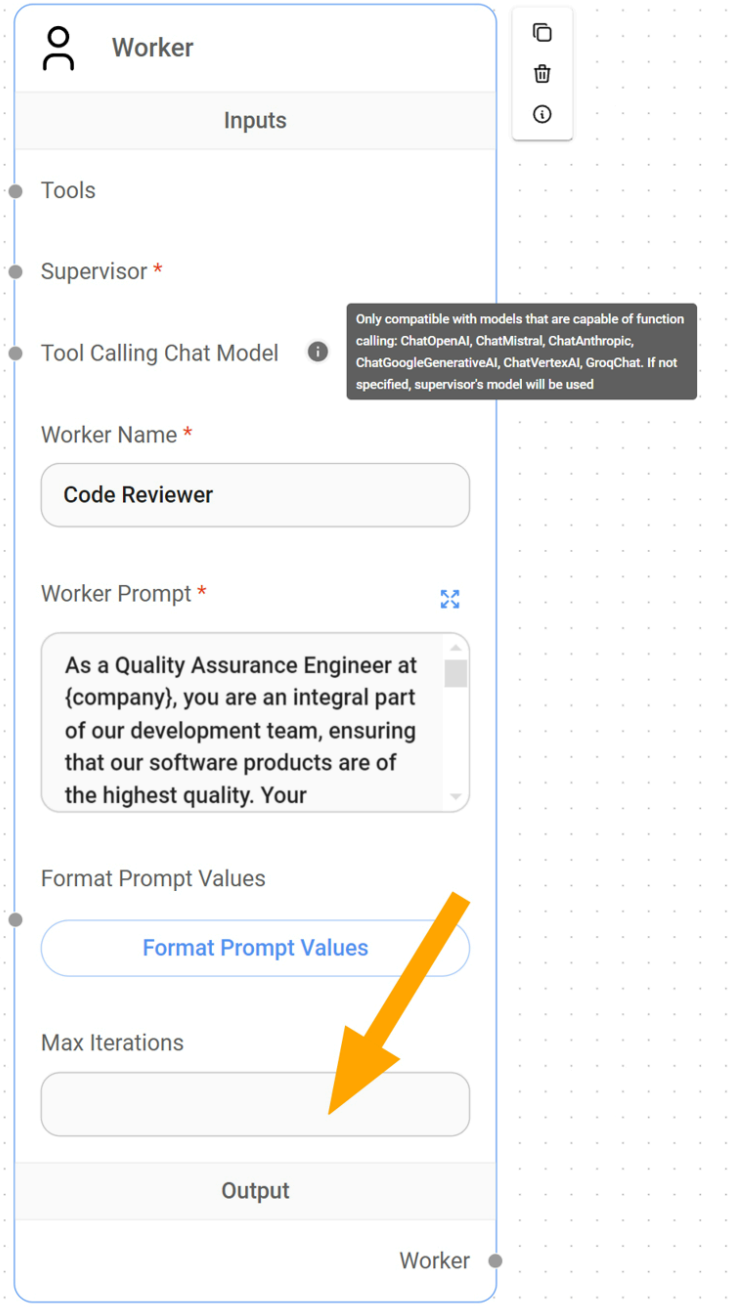
> Since the Supervisor only has the Workers' names to infer their functionality inside the workflow, it is very important that those names are set accordingly. **Clear, concise, and descriptive names that accurately reflect the Worker's role or area of expertise are crucial for the Supervisor to make informed decisions when delegating tasks.** This ensures that the right Worker is selected for the right job, maximizing the system's accuracy in fulfilling the user's request.

The Worker

The Worker, as a specialized agent instructed to handle a specific task within the system, requires two essential components to function correctly:

- **A Supervisor:** Each Worker must be connected to the Supervisor so it can be called upon when a task needs to be delegated. This connection establishes the essential hierarchical relationship within the multi-agent system, ensuring that the Supervisor can efficiently distribute work to the appropriate specialized Workers.

- **A Chat Model node capable of function calling**: By default, Workers inherit the Supervisor's Chat Model node unless assigned one directly. This function-calling capability enables the Worker to interact with tools designed for its specialized task.


Worker
Inputs
Tools
Supervisor *
Tool Calling Chat Model
Only compatible with models that are capable of function calling: ChatOpenAI, ChatMistral, ChatAnthropic, ChatGoogleGenerativeAI, ChatVertexAI, GroqChat. If not specified, supervisor's model will be used
Worker Name *
Code Reviewer
Worker Prompt *
As a Quality Assurance Engineer at {company}, you are an integral part of our development team, ensuring that our software products are of the highest quality. Your
Format Prompt Values
Format Prompt Values
Max Iterations
Output
Worker

> The ability to assign **different Chat Models to each Worker** provides significant flexibility and optimization opportunities for our application. By selecting Chat Models tailored to specific tasks, we can leverage more cost-effective solutions for simpler tasks and reserve specialized, potentially more expensive, models when truly necessary.

Undertanding Max Iteration parameter in Workers

LangChain refers to `Max Iterations Cap` as a important control mechanism for preventing haywire within an agentic system. In our current this context, it serves us as a guardrail against excessive, potentially infinite, interactions between the Supervisor and Worker.

Unlike the Supervisor node's `Recursion Limit`, which restricts how many times the Supervisor can call itself, the Worker node's `Max Iteration` parameter limits how many times a Supervisor can iterated or query a specific Worker.

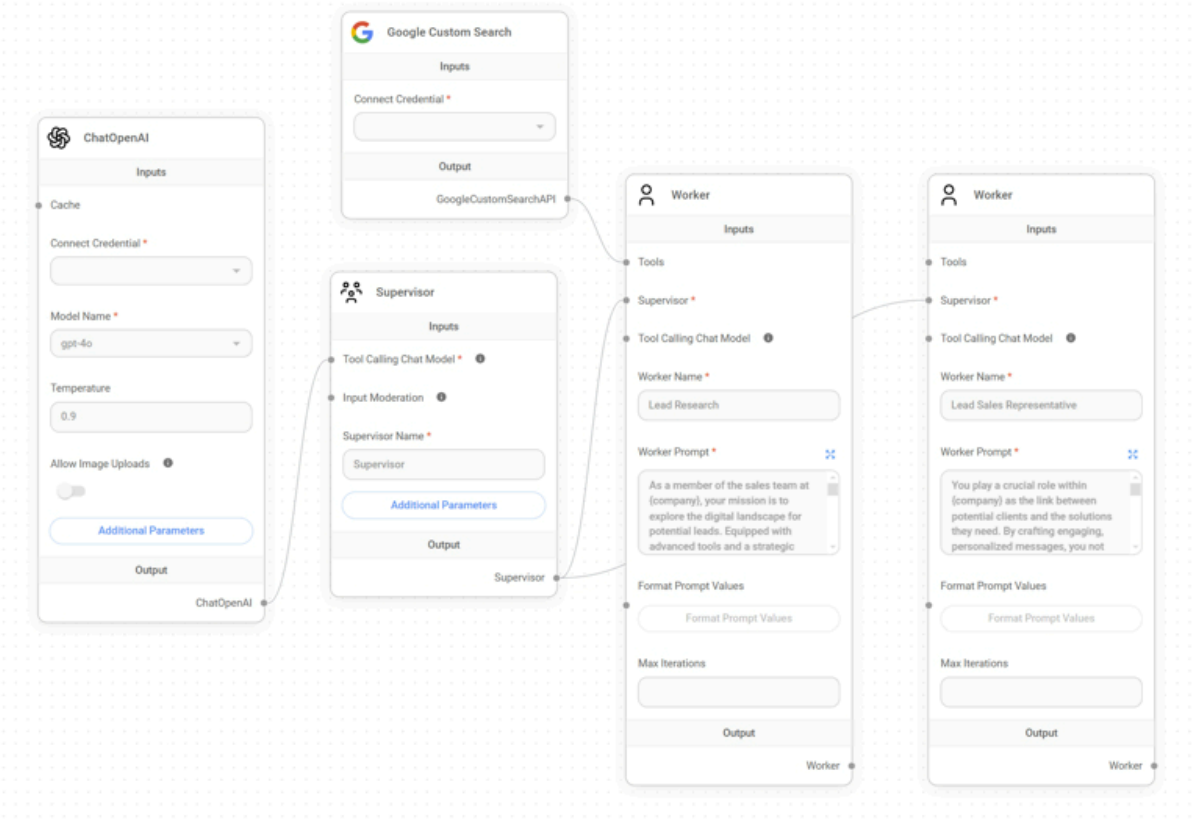
By capping or limiting the Max Iteration, we ensure that costs remain under control, even in cases of unexpected system behavior.

Example: A practical user case

Now that we've established a foundational understanding of how Multi-Agent systems work within Flowise, let's explore a practical application.

Imagine a **Lead Outreach multi-agent system** (available in the Marketplace) designed to automate the process of identifying, qualifying, and engaging with potential leads. This system would utilize a Supervisor to orchestrate the following two Workers:

- **Lead Researcher:** This Worker, using the Google Search Tool, will be responsible for gathering potential leads based on user-defined criteria.
- **Lead Sales Generator:** This Worker will utilize the information gathered by the Lead Researcher to create personalized email drafts for the sales team.



Background: A user working at Solterra Renewables wants to gather available information about Evergreen Energy Group, a reputable renewable energy company located in the UK, and target its CEO, Amelia Croft, as a potential lead.

User Request: The Solterra Renewables employee provides the following query to the multi-agent system: "*I need information about Evergreen Energy Group and Amelia Croft as a potential new customer for our business.*"

1. **Supervisor:**

- The Supervisor receives the user request and delegates the "Lead Research" task to the `Lead Researcher Worker`.

2. **Lead Researcher Worker:**
   - The Lead Researcher Worker, using the Google Search Tool, gathers information about Evergreen Energy Group, focusing on:
     - Company background, industry, size, and location.
     - Recent news and developments.
     - Key executives, including confirming Amelia Croft's role as CEO.
   - The Lead Researcher sends the gathered information back to the `Supervisor`.
3. **Supervisor:**
   - The Supervisor receives the research data from the Lead Researcher Worker and confirms that Amelia Croft is a relevant lead.
   - The Supervisor delegates the "Generate Sales Email" task to the `Lead Sales Generator Worker`, providing:
     - The research information on Evergreen Energy Group.
     - Amelia Croft's email.
     - Context about Solterra Renewables.
4. **Lead Sales Generator Worker:**
   - The Lead Sales Generator Worker crafts a personalized email draft tailored to Amelia Croft, taking into account:
     - Her role as CEO and the relevance of Solterra Renewables' services to her company.

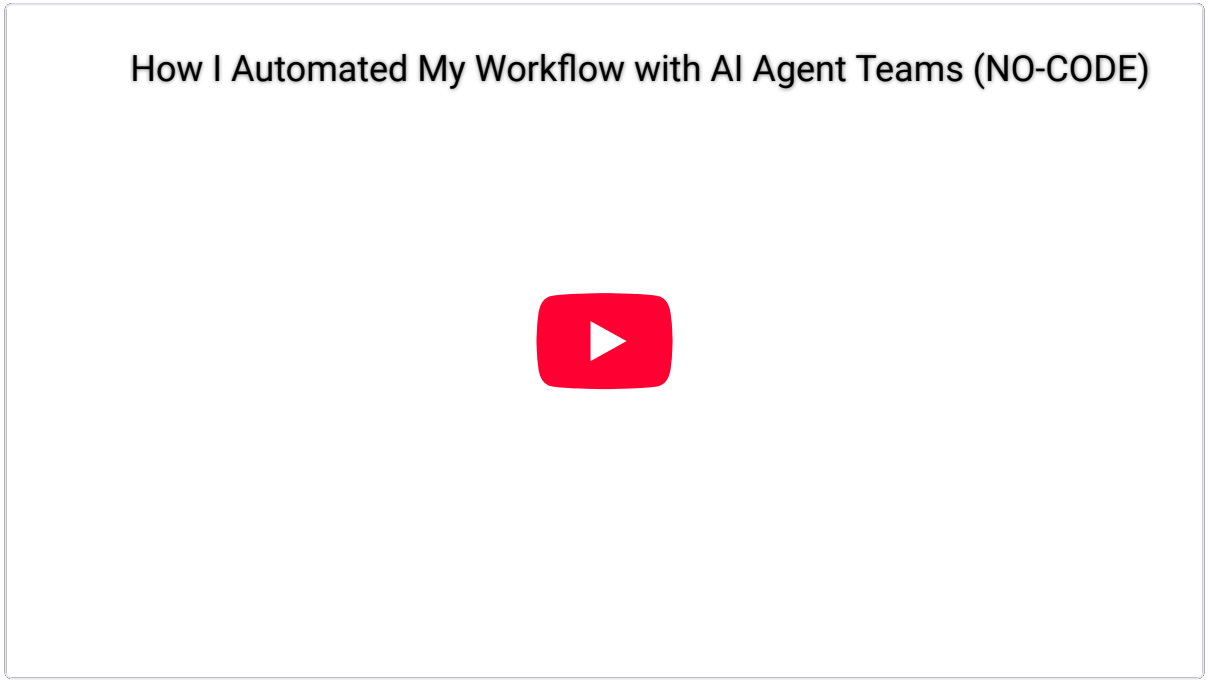
    - Information from the research about Evergreen Energy Group's current focus or projects.
  - The Lead Sales Generator Worker sends the completed email draft back to the `Supervisor`.
5. **Supervisor:**
  - The Supervisor receives the generated email draft and issues the "FINISH" directive.
  - The Supervisor outputs the email draft back to the user, the `Solterra Renewables employee`.
6. **User Receives Output:** The Solterra Renewables employee receives a personalized email draft ready to be reviewed and sent to Amelia Croft.

Video Tutorials

Here, you'll find a list of video tutorials from Leon's YouTube channel showing how to build multi-agent applications in Flowise using no-code.

How to Build AI Agent Teams (NO-CODE!) | Flowise Agentflows ...

Build Multi-Agent AI Teams with RAG & Custom Tools! (No-Code...



Previous
Agentflows

Next
Sequential Agents

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